

GLASGOW COLLEGE UESTC

Main Exam Paper

Power Electronics (UESTC3022)

Date: 8th June 2022

Time: 19:00-21:00

Attempt all PARTS. Total 100 marks

Use one answer sheet for each of the questions in this exam.

Show all work on the answer sheet.

Make sure that your University of Glasgow and UESTC Student Identification Numbers are on all answer sheets.

An electronic calculator may be used provided that it does not allow text storage or display, or graphical display.

All graphs should be clearly labelled and sufficiently large so that all elements are easy to read.

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

DATA/FORMULAE SHEET IS PROVIDED AT THE FRONT OF PAPER

Power Electronics 2 Formulae List

For an inductor: $v_L(t) = L \frac{di_L(t)}{dt}$, $i_L(t) = \frac{1}{L} \int v_L(t) dt$, $E_L(t) = \frac{1}{2} L [i_L(t)]^2$

For a capacitor: $v_C(t) = \frac{1}{C} \int i_C(t) dt$, $i_C(t) = C \frac{dv_C(t)}{dt}$, $E_C(t) = \frac{1}{2} C [v_C(t)]^2$

Average value: $V_{ave} = \frac{1}{T} \int_0^T v(t) dt$

RMS value: $V_{rms} = \sqrt{\frac{1}{T} \int_0^T v(t)^2 dt}$

Duty cycle: $D \equiv \phi = \frac{T_{on}}{T_{on} + T_{off}} = \frac{T_{on}}{T_{period}}$

Amplitude modulation index of SPWM:

$$m_a = \frac{\hat{V}_{control}}{\hat{V}_{carrier}}$$

Frequency modulation index of SPWM:

$$m_f = \frac{f_{carrier}}{f_{control}}$$

Instantaneous power:

$$p(t) = \frac{v^2(t)}{R} = i^2(t)R = v(t) \times i(t) = \text{force} \times \text{velocity} = \text{torque} \times \text{angular velocity}$$

Average power:

$$P = \int_0^T v(t)i(t)dt = \frac{V_{rms}^2}{R} = I_{rms}^2 R$$

reverse recovery power loss in a diode:

$$P_{RR} = Q_{RR} \times V_{R(blocking)} \times f_{switching}$$

MOSFET switching loss:

$$P_{switching} = \frac{f_{switching} V_{DS(off)}}{2} (T_{on} I_{on} + T_{off} I_{off})$$

Thermal circuit:

$$\Delta T = P_{device} \times \Sigma \theta_{XY} \quad \text{or} \quad \Delta T(^{\circ}\text{C}) = \frac{\text{power} \times \text{time}}{\text{thermal capacitance}}$$

Junction temperature of device:

$$T_{junction} = T_{sink} + P_{device} (\theta_{junction-case} + \theta_{case-sink})$$

Calculus:

$$\cos 2\theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\int \cos(a\theta) d\theta = \frac{1}{a} \sin \theta \quad \int \sin(a\theta) d\theta = -\frac{1}{a} \cos \theta$$

Q1. Answer all questions in this section:

a. For an inductor load $L=1\text{ mH}$, the steady-state inductor current waveform is shown in Figure Q1-a,

- i. Calculate the average current and RMS current for the inductor. [2]
- ii. Derive and sketch the corresponding inductor voltage, labelling the axes clearly. [3]

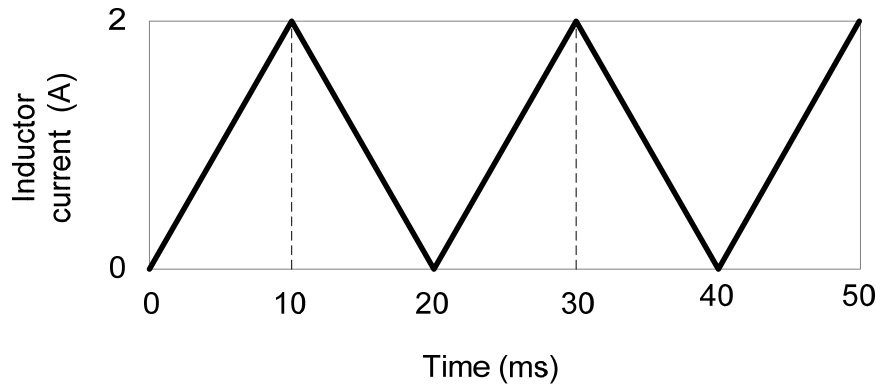


Figure Q1-a

b. Figure Q1-b shows the Forward Bias Safe Operating Area (FBSOA) of one IGBT.

- i. Please explain what is meant by the Safe Operating Area (SOA)? [1]
- ii. Please point out what do the four curve items ①, ②, ③, ④ stand for? [4]

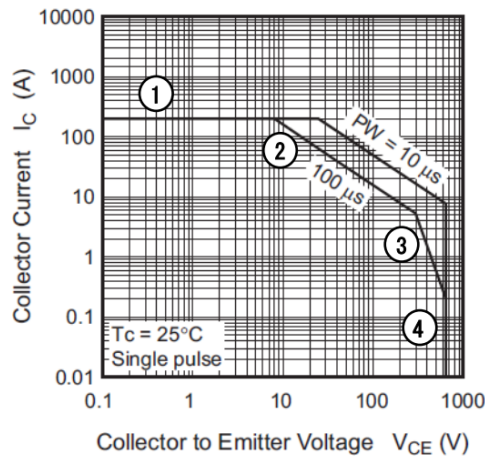


Figure Q1-b

c. Figure Q1-c shows the turn-on and turn-off transient voltage and current waveforms of a power diode. Please relate the labels ① to ⑤ to the following five parameters of the power diode:

- i. Forward voltage drop, V_F . [1]

- ii. Forward current, I_F . [1]
- iii. Peak forward voltage, V_{FP} . [1]
- iv. Maximum reverse voltage, V_{rr} . [1]
- v. Maximum reverse current, I_{rr} . [1]

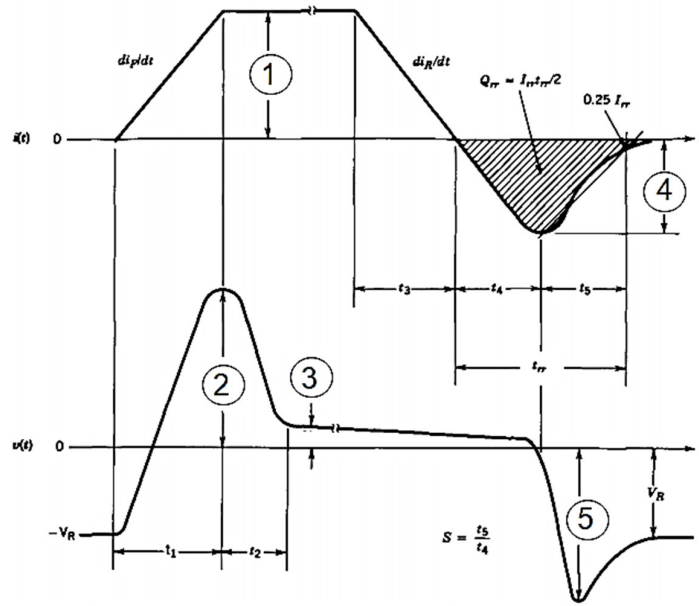


Figure Q1-c

- d. Figure Q1-d shows a thyristor rectifier T which is powered by the main voltage supply u_2 and switch-controlled gate voltage supply E_g .
- i. Please briefly describe the operating principle of the thyristor. [2]
 - ii. Please draw the waveform of the voltage drop, V_{load} , across the load resistor, R_d , and indicate the delay angle and the time scale clearly. [3]

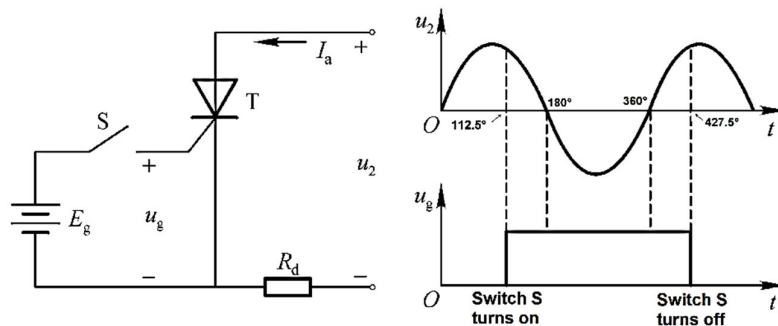


Figure Q1-d

- e. Consider the circuit in Figure Q1-e. Assume the power diodes D_1 and D_2 are ideal diodes, i.e., their forward voltage drop = 0 V. Determine the following when a 120 V rms sine voltage is applied to the primary winding of the transformer:

- What type of circuit is this? [1]
- Draw the voltage waveform across R_L . [2]
- What is the PIV for each diode? [2]

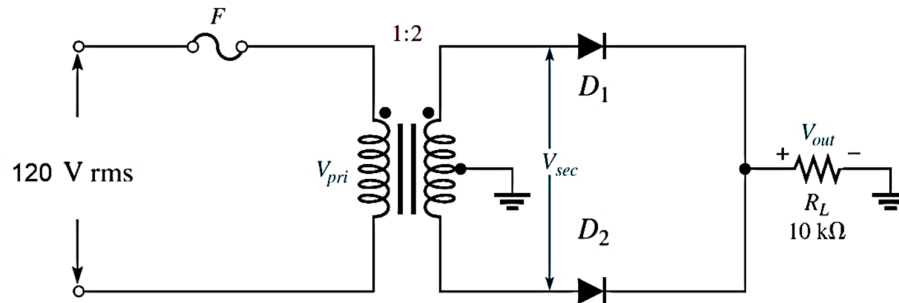


Figure Q1-e

- Q2. For the full-wave diode rectifier circuit shown in Figure Q2, assuming the input voltage is $V_{AC} = 60 \sin(100\pi t)$, the conduction angle of the diode is 30° , the load resistance $R_L = 10 \Omega$, and the capacitor $C_S = 15,000 \mu\text{F}$. We assume that all components are perfect and neglect the voltage drop across the diodes.
- Estimate the peak-to-peak ripple voltage in the output voltage, V_R , and the peak inverse voltage in the voltage drop, V_D , for the diode, D . [5]
 - If the output filter capacitor, C_S , is removed, sketch and label the voltages across the load, R_L , and the diode, D , separately, showing the time scale clearly. Calculate the RMS value of the output voltage, V_R . [5]
 - If the ripple voltage across the load resistance, R_L , must not exceed 1 V peak-to-peak. Calculate the value of smoothing capacitance, C_S . You may assume that the capacitor provides current to the load for each complete cycle of the rectified voltage. [5]
 - If only increase the value of C_S , what will it affect V_R ripple, the power diode conduction angle, the form factor of the current waveform drawn from the transformer, and the VA rating of the transformer, respectively? [5]
 - List the advantages and disadvantages of this full bridge rectifier circuit. [5]

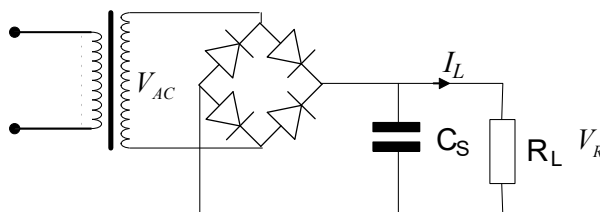


Figure Q2

Q3. A switched-mode buck converter is required to deliver 5.0 V at 500 mA from a 12.0 V supply. It runs in continuous conduction mode at 50 kHz. The circuit is shown in figure Q3 and has a 100 μH inductor. You may assume that all components are ideal and that the input and output are smoothed so effectively that their voltages may be treated as constant.

- Calculate the duty cycle required for the converter. [2]
- Calculate the average input current. [2]
- What are the voltages across the inductor during charging and discharging? [2]
- What is the value of the average current through the inductor? [1]
- Calculate the values of di_L/dt during charging and discharging. [3]
- Hence calculate the total change in i_L during charging and confirm that it falls by the same value during discharging. [4]
- What are the minimum and maximum values of $i_L(t)$? [2]
- Sketch $i_L(t)$, $i_{in}(t)$, $i_{out}(t)$ and $v_L(t)$. An accurate scale drawing is not required but you must show the numerical values at all important points. [6]
- Suppose that a linear regulator was chosen for the supply instead. What would be its maximum efficiency and minimum power dissipation? [3]

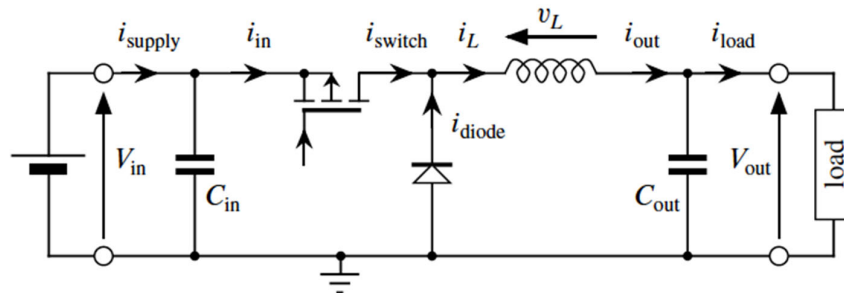


Figure Q3

Q4. In the circuit shown in Figure Q4, the switch operates at 100 kHz, the supply voltage is 200 V, and the forward voltage drop of any diode is 1 V. With this in mind, answer the following questions

- What is the snubber circuit for? [4]
- If the peak current through the switch is 2 A:
 - What power is dissipated in the resistor? [3]

- (ii) Referring to figure Q4, If the voltage at point A is 200 V, what is the rate of change (?) of current limited to when the switch is closed? [3]
- (iii) What is the peak voltage across the inductor and switch? [2]
- (iv) Assuming the switch in Figure Q4 has an output capacitance of 300 pF and the stray inductance due to circuit board layout is 800 nH. What is the undamped ringing frequency? [4]

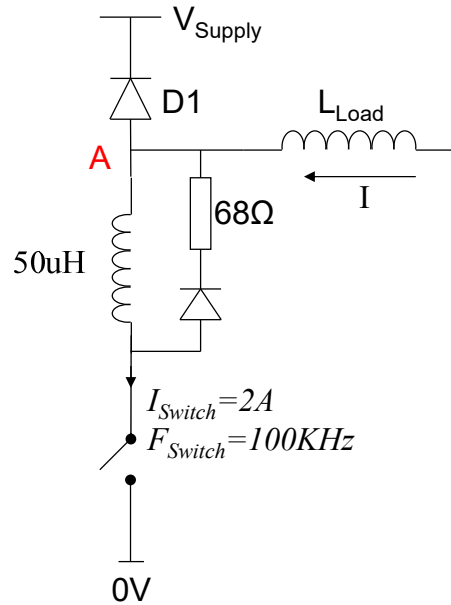


Figure Q4

- c. A power diode has a thermal resistance θ_{JC} of $0.1\text{ }^{\circ}\text{C/W}$. The diode is bolted to a heatsink of thermal resistance $0.15\text{ }^{\circ}\text{C/W}$, and the thermal resistance of the contact between the diode case and the heatsink is $0.04\text{ }^{\circ}\text{C/W}$. If the diode is operating in free air with an ambient temperature of $25\text{ }^{\circ}\text{C}$,
 - (i) Draw the thermal circuit diagram of the diode. [3]
 - (ii) What is the heatsink temperature, case temperature and junction temperature when the diode is dissipating 200 W? [6]